Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with electrochemical materials, DOE-guided process development, metrology-style characterization, and failure-analysis experience across polymer and thin-film systems. Builds structure-property datasets and reproducible test workflows relevant to advanced battery-materials concept development.

CORE COMPETENCIES

Battery Materials Adjacent | Metrology & Characterization | Failure Analysis | DOE/SPC/JMP | Electrochemical Materials | Structure-Process-Property | Process Development | Technical Documentation

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided process-development experiments for bio-based polymer systems, connecting processing conditions, characterization readouts, and reproducible specifications for scale-up.
- Use spectroscopy, FT-IR, and thermal analysis to build structure-property datasets and identify process parameters that drive material performance.
- Create SOP-ready documentation and safety-conscious lab workflows that improve experimental repeatability and technology-transfer readiness.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Designed and evaluated 15+ electrochemical material variants using cyclic voltammetry, spectroelectrochemistry, UV-Vis-NIR, FT-IR, TGA, DSC, NMR, and GPC to map structure-property-performance relationships.
- Identified material reliability risks including oxidative degradation, interface delamination, and redox instability, applying root-cause analysis to recommend design and process mitigations.
- Built quantitative datasets from optical, electrochemical, and thermal characterization to guide material selection and failure-analysis decisions.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded thin-film/coating process development, using surface preparation and CVD process controls to improve film morphology and device performance.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Developed QCM-D-monitored polymer thin-film deposition workflows and DOE-based process optimization to improve coating reproducibility and uniformity.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed operations for a 15-member process-development laboratory, implementing equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30%.
- Trained researchers on instrument workflows, experimental documentation, and safe handling while coordinating multiple concurrent materials-development projects.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Electrochemical Materials Structure-Property Analysis — Battery-adjacent Materials

Designed and characterized electrochemical material variants to connect molecular design, redox behavior, optical response, and degradation mechanisms.

Cyclic voltammetry, spectroelectrochemistry, UV-Vis-NIR, FT-IR, TGA, DSC

DOE-Guided Process Development — Concept Development

Led polymer process experiments that define processing windows, quality readouts, and reproducible documentation for scale-up-facing materials workflows.

DOE, SPC, FT-IR, thermal analysis, SOPs

Materials Reliability and Failure Modes — Failure Analysis

Investigated oxidative degradation, delamination, and redox instability, applying root-cause analysis to recommend material and process mitigations.

Stress testing, spectroscopy, electrochemistry, root-cause analysis

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder — 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) — 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Materials & Battery-Adjacent: Electrochemical materials, polymer systems, thin films, coatings, structure-property relationships, process-property analysis
Metrology & Characterization: Cyclic voltammetry, UV-Vis-NIR, FT-IR, TGA, DSC, NMR, GPC, QCM-D, spectroelectrochemistry
Process & Quality: DOE, SPC, JMP/basic, process windows, reproducibility studies, SOPs, calibration systems, safety protocols
Reliability & Data: Failure-mode identification, root-cause analysis, degradation/stability testing, technical reports, Excel, Python/basic

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)